Mapping the landscape of music artist similarity research

The most promising unexplored territory for capturing "invisible lines" between artists lies in three directions: mining artist self-discourse (interviews, stated philosophy), formally measuring the orthogonality between independent similarity axes, and exploiting enthusiast-curated platforms like RateYourMusic as a distinct collaborative signal. Standard approaches — collaborative filtering, audio embeddings, tag-based methods, and playlist co-occurrence — are mature and well-documented across hundreds of papers. But a systematic gap persists: no existing system combines the full spectrum of available signals, and several rich data sources remain entirely untapped. The 2023–2026 period has seen a decisive shift toward foundation models (CLAP, MuQ-MuLan) and LLM-based recommendation, with cross-modal contrastive learning emerging as the most active frontier. Below is a comprehensive map of what exists, what's genuinely novel, and where the field is heading.

The established toolkit is deep but siloed

Eight major approaches to music artist similarity have been explored extensively, each with a substantial body of literature. They can be grouped by signal type:

Collaborative filtering and its modern variants remain the industry backbone. Standard matrix factorization on implicit feedback (play counts, skips) powers Spotify and Last.fm. Graph neural networks — particularly **LightGCN, PinSage, and GraphSAGE** — have emerged as powerful extensions, combining interaction topology with content features. Korzeniowski, Oramas, and Gouyon's 2022 paper in *Transactions of ISMIR* introduced the OLGA dataset (**17,673 artists** with AllMusic similarity ground truth and AcousticBrainz audio features) and demonstrated GNNs with triplet loss and "connection dropout" regularization for long-tail artists. Transformer-based sequential models (SASRec, BERT4Rec) have been applied to music but face a unique challenge: repetitive consumption patterns mean that a simple "Personalized Most Popular" baseline often outperforms them, as shown in 2024 arxiv studies.

Audio content-based similarity has evolved from MFCCs through deep CNN embeddings (Van den Oord et al.'s landmark 2013 NIPS paper predicting CF latent factors from mel-spectrograms) to today's foundation models. CLAP (Contrastive Language-Audio Pretraining) and **MuQ-MuLan** (Tencent AI Lab, January 2025) represent the current state of the art, achieving **72.4% accuracy** on perceptual music similarity benchmarks. An ETH Zurich study (Grötschla et al., 2024) directly tested CLAP embeddings for artist-level similarity on the OLGA dataset, finding them effective and complementary to graph topology.

Text and NLP approaches trace back to Whitman and Lawrence's foundational 2002 paper at MIT Media Lab, which built TF·IDF term profiles from web pages about artists. The key evolution came with Oramas et al.'s 2015 ISMIR paper, which applied entity linking (via Babelfy) to **Last.fm biographies**, connecting extracted entities to BabelNet/DBpedia for semantic enrichment — significantly outperforming surface-level text approaches. Logan et al. (2004) explored lyrics via Probabilistic Latent Semantic Analysis on **40,000+ songs**.

Playlist co-occurrence is arguably the strongest signal in production systems. The Spotify Million Playlist Dataset (1 million playlists, 2+ million tracks) drove the ACM RecSys 2018 Challenge, where winning approaches used two-stage architectures: high-recall candidate retrieval via neighborhood CF, followed by neural re-ranking. Word2Vec-style embeddings treating playlists as "sentences" and tracks as "words" proved surprisingly effective.

Knowledge graph approaches, **tag-based/folksonomy methods** (Last.fm tags analyzed extensively by Lamere in 2008), **social media mining** (Schedl's pioneering 2010 work on Twitter #nowplaying data), and **influence/lineage networks** (Bryan and Wang's 2011 ISMIR analysis of WhoSampled data; Figueiredo and Andrade's 2019 AllMusic influence graph) round out the established approaches. Each captures a distinct facet of artist relationships, but they have rarely been combined comprehensively.

The HDSR influence graph paper is the closest precedent

The most directly relevant prior work to the "invisible lines" concept is Elena Badillo-Goicoechea's 2025 paper in *Harvard Data Science Review* (Issue 7.4), "Modeling Artist Influence for Music Selection and Recommendation: A Purely Network-Based Approach." Working from the University of Chicago's Biological Sciences Division, Badillo-Goicoechea constructed a knowledge graph of **22,831 artists connected by 159,389 edges** by mining critic namedrops from publications including Pitchfork, The Quietus, and NPR's All Songs Considered. Named entity recognition (NLTK + TextBlob) identified artist mentions in reviews; co-mentions formed graph edges interpreted as influence signals.

The paper implements four graph traversal strategies — BFS, Dijkstra shortest path, and max-flow methods — to generate recommendations. Its most compelling output is influence chains connecting seemingly disparate artists: **The Velvet Underground → Cate Le Bon → St. Vincent → Taylor Swift**, or **Kraftwerk → Iggy Pop → Nine Inch Nails**. Simulation results showed the system matches collaborative filtering for general users and outperforms it for high-openness listeners. This is the first published system to explicitly target "bridge artist discovery" — finding artists that connect different musical worlds through the connective tissue of critical discourse.

The key limitation is that it relies entirely on third-party discourse (what critics write about artists), not on what artists say about themselves. It also does not combine its network signal with audio features in the graph construction phase, though it uses Spotify audio features for optional post-hoc sonic similarity weighting.

Seven specific approaches and their novelty status

A systematic search across academic databases, ISMIR/RecSys proceedings, arxiv, and independent projects reveals a clear gradient from "entirely unexplored" to "moderately studied" for the candidate novel approaches:

Artist self-discourse as a similarity signal is genuinely untapped. No published work uses what artists say about their own creative process — from interviews, podcasts, press conferences, or artist statements — as features for recommendation or similarity computation. The entire NLP-for-music-similarity literature (Whitman & Lawrence 2002 through Oramas et al. 2015) uses text written *about* artists by third parties. The closest conceptual neighbor is sentiment analysis of lyrics (the artist's creative output, not their meta-discourse about it). The raw data exists in abundance — interview transcripts, podcast databases, artist quote collections — but has never been systematically extracted and used as a similarity signal. This represents a **genuine research contribution** with no direct precedent.

Formal orthogonality measurement between similarity axes is absent. Ellis, Whitman, Berenzweig, and Lawrence's 2002–2003 Columbia papers ("The Quest for Ground Truth in Musical Artist Similarity") compared acoustic, collaborative, and editorial similarity measures, finding moderate agreement — but stopped short of formally measuring orthogonality. Bogdanov et al. combined timbral, tempo, and semantic distances, showing hybrids outperform individuals (implying complementarity) without quantifying independence. Schedl et al.'s 2013 survey predicted "multi-faceted similarity measures will soon be standard" — but **12 years later, no paper has explicitly measured how much independent information each axis provides**. Quantifying the information geometry of music similarity space would be a novel contribution. This sounds promising and I'm into it, but it'll take some real digging into to make it tangible.

WhoSampled data has been analyzed but never built into a recommender. Bryan and Wang's 2011 ISMIR paper analyzed sampling networks for musicological insight (revealing funk/soul/disco's dominance as source material for hip-hop). A 2024 arxiv paper applied Bayesian Surprise to temporal evolution of influence networks. Spotify's **acquisition of WhoSampled in November 2025** signals industry recognition of this signal's value. But no formal recommender system using sampling relationships as a primary signal has been published — the gap between network analysis and actionable recommendation remains open.

Could be useful, but not exactly what I'm looking to measure

Production technique similarity exists at the margins. The "Producer Effect" is a well-documented confound in MIR — audio classifiers often learn production characteristics (EQ, compression, stereo width) rather than musical content. A 2022 arxiv paper on "Music Mixing Style Transfer" trained an FXencoder with contrastive learning to capture mixing style disentangled from musical content. Stereo panning features have been proposed for classifying recording production style. But **no recommendation system explicitly uses production technique as a named feature** — it remains a confound to be controlled rather than a signal to exploit. Could be a good feature for a different project—though I agree it might add confounds to mine

RateYourMusic and Bandcamp collection data are academically untouched. RYM has its own internal recommendation system using 5-star ratings, but no published MIR research uses RYM or Bandcamp data. The vast majority of collaborative filtering research relies on Last.fm, the Million Song Dataset, or Spotify — all streaming-based platforms where interaction signals are passive. RYM users are self-selecting enthusiasts who actively rate and curate, producing higher-signal preference data. RYM's rich themed list culture — curated lists like "Best shoegaze albums with female vocalists" — represents an entirely untapped signal distinct from playlist co-occurrence.

RYM database IS great but locked away/API has been an empty promise. Scraping would be the only way

Album liner note text has never been used for NLP similarity. MusicBrainz captures structural credit data (personnel, producers, engineers), and this has been used implicitly for personnel-overlap networks. But the textual content of liner notes — essays, artist statements, philosophical commentary about creative intent — has never been processed as NLP features for similarity. The WASABI Song Corpus combines metadata with NLP-processed lyrics but excludes liner notes. Digitization challenges partially explain this gap. I'll have to look at some examples. Dataset might be small/heavily artist dependent.

Temporal convergence between artist trajectories is nascent. Collins's 2025 paper in *Royal Society Open Science* tracked artist career trajectories using timbral-rhythmic and harmonic features, comparing R.E.M., Radiohead, and Coldplay to find inter-artist similarity evolving over careers. Mauch et al.'s 2015 study tracked how musical features evolved across the Billboard Hot 100 from 1960–2010. Georges et al. discussed "convergent evolution" in classical music. But the specific task of recommending artists based on trajectory convergence — "their sound is evolving toward yours" — has not been operationalized.

Also would need some real fleshing out to make a formalized approach.

The 2023–2026 frontier is defined by LLMs and cross-modal fusion

Three developments are reshaping the field. First, **LLMs are becoming music recommendation engines**. The definitive survey by Epure, Deldjoo, Sguerra, Schedl, and Moussallam (November 2025, arxiv) identifies two paradigms: LLM as ranker (scoring candidates) and LLM as embedding model (mapping users and items into shared latent space). Spotify has been the most aggressive industrial player, introducing "Semantic IDs" (De Nadai et al., November 2025) — a vocabulary that lets LLMs refer to specific catalog items as tokens, enabling the model to "think" in terms of musical entities. TalkPlay (Doh and Nam, ICML 2025) reformulates music recommendation as next-token prediction, unifying dialogue, retrieval, and ranking in a single generative model. FusID (January 2026) improves on this with modality-fused semantic IDs, achieving **6.21% MRR improvement** over TalkPlay.

Second, **cross-modal contrastive learning is the most active research frontier**. CrossMuSim (Huawei, ICASSP 2025) uses GPT-4o-mini and Qwen2-7B to generate rich text descriptions from just artist name and song title, then trains audio embeddings via contrastive learning against these descriptions. Deployed on Huawei Music via A/B testing, it showed **15.57% improvement** over baselines for Cantopop and significant gains across J-pop and K-pop. A critical finding: the music encoder internalizes strong artist-specific features (timbre) that persist even when artist names are removed from training text — suggesting audio models naturally learn artist-level signatures. JAM (Just Ask for Music, RecSys 2025) models user-query-item interactions as vector translations in a shared latent space inspired by TransE knowledge graph embeddings.

Third, **heterogeneous graph neural networks for artist similarity** have matured significantly. Da Silva et al.'s August 2024 paper in *IEEE/ACM Transactions on Audio, Speech and Language Processing* proposed the first multimodal heterogeneous graph integrating audio, lyrics, and artist relations for artist similarity via link prediction. GATSY (Sapienza/ISTI-CNR, 2024–2025) introduced "fictitious artists" as bridge nodes to connect previously unlinked artists — and found that **genre-based supervision does not improve artist similarity**, validating that invisible lines transcend genre boundaries. MUSYNERGY (2025) goes beyond similarity to predict potential creative partnerships using heterogeneous knowledge graphs from MusicBrainz.

Notable non-academic projects push boundaries too. **Cosine.club** (April 2024) uses Essentia/UPF's Discogs-EfficientNet model to generate embeddings for **1.15 million electronic music tracks**, returning unknown artists that share audio DNA with queries — deliberately breaking the popularity bias of collaborative filtering. **MusicLynx** (Allik et al.) discovers non-obvious connections through shared metadata: ethnic background, nationality, record label, "shared fate or affliction." **Music Galaxy** by Casey Primozic visualizes 70,000+ artists in 3D using node2vec embeddings of Spotify's Related Artists graph.

Where the genuine white space lies

Synthesizing across all sources, the most impactful research directions for an "invisible lines" project fall into three tiers:

The **highest-novelty opportunities** are artist self-discourse mining (no precedent whatsoever), formal orthogonality measurement between similarity axes (acknowledged as needed since 2002 but never done), and RYM/Bandcamp enthusiast-curated data as a distinct collaborative signal (academically untouched despite clearly different demographic and signal properties from streaming CF).

Medium-novelty opportunities with partial precedent include WhoSampled-based recommendation (analysis exists, system does not), liner note text NLP (structural credits used but textual content ignored), production technique as explicit feature (confound identified but never exploited), and temporal trajectory convergence as recommendation signal (Collins 2025 is the sole direct precedent).

The **key architectural insight** from the survey literature is that despite Deldjoo et al.'s 2024 "onion model" identifying five layers of music content (signal, embedded metadata, expert-generated content, user-generated content, derivative content), **no system has been demonstrated that fuses all signal types simultaneously** for artist-level similarity. Most hybrid approaches combine two or three signals. A system that quantifies the independent contribution of each signal — measuring orthogonality — while combining them would be structurally novel.

Conclusion

The music artist similarity landscape is simultaneously mature and riddled with gaps. The mature end — collaborative filtering, audio embeddings, playlist co-occurrence — powers billion-dollar streaming platforms. The gap end includes entire categories of data that have never been touched: artist self-discourse, liner note essays, enthusiast-curated collections, production technique metadata. The most significant conceptual gap is that no one has formally measured how independent different similarity signals are from each other — a prerequisite for knowing which "invisible lines" each approach can uniquely reveal. The 2023–2026 shift toward LLMs and cross-modal foundation models creates new infrastructure for closing these gaps: an LLM can process interview transcripts, liner notes, and reviews simultaneously, while contrastive learning can align these heterogeneous text signals with audio embeddings in a shared space. The GATSY finding that genre supervision *hurts* artist similarity performance is perhaps the single most important recent validation of the "invisible lines" thesis — the connections that matter most are precisely the ones that genre labels obscure.